Product Requirements Document (PRD): Enterprise AI Meeting Intelligence Platform

1. Executive Summary

This document outlines the requirements for a high-performance, enterprise-grade AI Meeting Intelligence Platform. The system leverages a multi-agent "Trunk & Bus" architecture to provide real-time transcription, context-aware memory, and proactive automation for Google Meet. By utilizing Gemini 2.5 Flash and a specialized agent layer, the platform transforms live conversations into actionable structured data while maintaining rigorous security and observability standards.

2. Problem Statement

Professional meetings are often information-dense, leading to lost context, inefficient manual note-taking, and delayed follow-ups. Late joiners lack immediate context, and historical knowledge across multiple meetings is difficult to retrieve. Existing solutions often lack real-time reasoning, enterprise-grade security, and deep integration into existing workflows (Slack, Gmail, Calendar).

3. Goals & Objectives

- Zero-Latency Intelligence: Provide real-time transcription and insights during the meeting.
- Agentic Memory: Enable reasoning over historical meeting data using vector embeddings.
- Proactive Automation: Automate post-meeting workflows (emails, Slack updates, calendar reminders) immediately upon meeting conclusion.
- Enterprise Compliance: Ensure 100% observability, GDPR compliance, and data encryption.

4. Target Users / Stakeholders

- Enterprise Professionals: Primary users seeking to automate meeting overhead.
- Project Managers: Users requiring accurate tracking of decisions and action items.
- IT/Security Teams: Stakeholders responsible for data privacy, compliance, and system observability.

5. Functional Requirements

ID	Feature	Description	Priority
FR-01	Real-time Transcription	DOM-based extraction of Google Meet captions with <1s latency.< /td>	Must
FR-02	Consent Capture	UI component to capture explicit participant consent before recording/transcribing.	Must
FR-03	Late Join Recap	On-demand summary of missed content for participants joining late.	Must
FR-04	Real-time Insights	Continuous stream (10s intervals) of key points and live sentiment.	Must
FR-05	Historical Memory	Semantic search and Q&A across historical meeting transcripts via Qdrant.	Must
FR-06	Data Encryption	AES-256 encryption at rest for all vector and transcript data.	Must
FR-07	Automated Erasure	Workflows for data minimization and right-to-erasure compliance.	Must
FR-08	Multi-Channel Sync	Automated drafting of follow-up emails and Slack notifications.	Should
FR-09	Calendar Sync	Create reminders and update event metadata using Google Calendar API.	Should

6. Non-Functional Requirements

- Observability: 100% of LLM calls must be traced via Langfuse and OpenTelemetry, including token usage, latency, prompt/version metadata, and correlation IDs.
- Tracing: All agent requests and LLM calls must propagate correlation IDs across Orchestrator, A2A Task Bus, Security Gateway, and observability pipeline.
- Budgeting: Each agent must enforce token and latency budgets, with per-call max token caps and alerting on budget breaches.
- Performance: Real-time insights must be delivered with a maximum 10-second update frequency.
- Scalability: Horizontal scaling of stateless agents on GKE; Qdrant collection partitioning for multi-tenant isolation.
- Reliability: Use of gRPC with backpressure and rate limiting to handle high-concurrency task loads.

7. System Architecture Overview

The system follows a "Trunk & Bus" architecture:

1. Frontend Layer: Chrome Extension (Sidebar) and Web Dashboard.
2. Security & Orchestration: Kong Gateway acts as the entry point, feeding into a "Security Trunk" that fanned out to agents.
3. Agent Layer: Seven specialized agents (Transcription, Insights, Recap, Memory, Email, Slack, Calendar) managed by a central Orchestrator using the Google A2A (Agent-to-Agent) protocol.
4. Communication: Asynchronous gRPC Task Bus for worker agents and a horizontal OTLP Trace Bus for observability.

5. Reasoning & Data: Dedicated LLM Layer (Gemini 2.5 Flash) and Vector Memory (Qdrant).

8. Tech Stack

- Frontend: TypeScript, Next.js, Tailwind CSS, shadcn/ui, Chrome Extension API.
- Orchestration: Python, Google ADK, Lyzr, LangGraph, Google A2A Protocol.
- Infrastructure: Kong Gateway (OAuth2, JWT, TLS 1.3), gRPC, OpenTelemetry (OTLP).
- AI/ML: Gemini 2.5 Flash.
- Database: Qdrant (Vector Search) with AES-256 encryption.
- Observability: Langfuse.

9. Data Requirements

- Vector Storage: Qdrant handles high-dimensional embeddings of meeting segments.
- Multi-tenancy: Data isolation enforced at the collection level.
- Data Flow: Transcripts flow from the DOM -> Orchestrator -> Security Trunk -> Specialized Agents -> Qdrant/External APIs.

10. API Specifications

- Internal: gRPC for high-performance, asynchronous A2A communication.
- External:
  - Gmail API: For drafting/sending follow-ups.
  - Slack API: For channel-based notifications.
  - Google Calendar API: For event metadata and reminders.
- LLM: Gemini 2.5 Flash API via secured LLM Layer.

11. Security Requirements

- Authentication: OAuth2 + JWT via Kong Gateway.
- Encryption: TLS 1.3 for data in transit; AES-256 for data at rest (Qdrant).
- GDPR Compliance: System must enforce explicit consent, data minimization, right to erasure, and automated retention/deletion workflows for

- transcript data.
- Input Validation: Strict schema validation at the Gateway level to prevent injection attacks.

12. Deployment & Infrastructure

- Cloud: Google Cloud Platform (GCP).
- Containerization: Kubernetes (GKE) for horizontal scaling of agents.
- CI/CD: Automated pipelines for agent deployment and Langfuse trace versioning.

13. Success Metrics

- Accuracy: Grounded recap accuracy above 95%, measured against provided transcript evidence.
- Latency: Transcription-to-Insight latency < 10 seconds.
- Reliability: 99.9% uptime for the A2A Task Bus.
- Compliance: 100% audit trail for data deletion requests.

14. Timeline & Milestones

- Phase 1: Core Transcription & Consent UI (MVP).
- Phase 2: Agent Orchestrator & Security Trunk implementation.
- Phase 3: Historical Memory (Qdrant) & Real-time Insights.
- Phase 4: External Integrations (Slack, Email, Calendar) & Observability Bus.

15. Open Questions & Risks

- Hallucination Risk: Mitigated by strict grounding requirements in agent prompts.
- API Rate Limits: Managed via Kong Gateway rate limiting and gRPC backpressure.
- DOM Stability: Google Meet UI changes may require frequent updates to the transcription extraction logic.

Appendix C: Late Join Recap Agent Specification

Framework: CRISPE (Capacity, Role, Insight, Statement, Personality, Experiment)

Prompt Constraint: “The agent must summarize only from the provided transcript segments. No hallucinations or external knowledge allowed.”

JSON Output Schema:

```
{
  "recap_summary": "string",
  "key_decisions": [
    {"decision": "string", "context": "string"}
  ],
  "action_items": [
    {"task": "string", "owner": "string", "deadline": "string"}
  ],
  "open_questions": ["string"],
  "risks": ["string"]
}
```

Few-Shot Example: Input: “Transcript: [00:01] Sarah: We decided to move the launch to Friday. [00:02] Bob: I'll handle the server migration by Thursday night.” Output:

```
{
  "recap_summary": "The team discussed the launch schedule and infrastructure readiness.",
  "key_decisions": [
    {"decision": "Launch moved to Friday", "context": "Agreed upon by Sarah"}
  ],
  "action_items": [
    {"task": "Server migration", "owner": "Bob", "deadline": "Thursday night"}
  ],
  "open_questions": [],
  "risks": ["Tight turnaround for server migration before Friday launch"]
}
```